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Name	Date modified	Type	Size
 a32_startup	7/11/2023 4:15 PM	File folder	
 build	7/7/2023 7:51 AM	File folder	
 drivers	7/7/2023 7:51 AM	File folder	
 example	7/11/2023 4:15 PM	File folder	
 include	7/11/2023 6:01 AM	File folder	
 lib	7/7/2023 7:51 AM	File folder	
 RTE	7/11/2023 4:15 PM	File folder	
 services_lib	7/11/2023 4:15 PM	File folder	
 A32_MRAM	7/11/2023 4:15 PM	Windows Script C...	
 A32_TCM	7/11/2023 4:15 PM	Windows Script C...	
 CMakeLists	7/11/2023 4:15 PM	Text Document	
 CMakePresets	7/11/2023 4:15 PM	JSON File	
 gcc_A32_MRAM	7/11/2023 4:15 PM	LD File	
 gcc_A32_TCM	7/11/2023 4:15 PM	LD File	
 gcc_M55_HE_MRAM	7/11/2023 4:15 PM	LD File	
 gcc_M55_HE_TCM	7/11/2023 4:15 PM	LD File	
 gcc_M55_HP_MRAM	7/11/2023 4:15 PM	LD File	
 gcc_M55_HP_TCM	7/11/2023 4:15 PM	LD File	
 License	7/7/2023 7:51 AM	Text Document	
 M55_HE_MRAM	7/11/2023 4:15 PM	Windows Script C...	
 M55_HE_TCM	7/11/2023 4:15 PM	Windows Script C...	
 M55_HP_MRAM	7/11/2023 4:15 PM	Windows Script C...	
 M55_HP_TCM	7/11/2023 4:15 PM	Windows Script C...	
 Makefile_linux	7/7/2023 7:51 AM	File	
 README	7/11/2023 4:15 PM	Markdown Source...	
 toolchain-armclang	7/11/2023 4:15 PM	CMake Source File	
 toolchain-gnu	7/11/2023 4:15 PM	CMake Source File	



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l c l f f t

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R r c r f f t

```
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2

[SES] es0 PPU PPU_PWRP=0x100 PPU_PWSR=0x108
[SES] es0 ppu_isr=0x80
[SES] es0 ppu_aISR=0x2
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fft fft fft r ftc l c c A

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fft fft fft r ftc l c c RA

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r f f t c c

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```
$ cmake .. -G "Unix Makefiles" -DENSEMBLE_CORE=M55_HE -DCMAKE_TOOLCHAIN_FILE=toolchain-armclang.cmake -DINSTALL_DIR=../junk
-- [INFO] Version=9
-- [INFO] installation override, using ../junk
-- The C compiler identification is ARMClang 6.18.2
-- The CXX compiler identification is ARMClang 6.18.2
-- The ASM compiler identification is ARMClang
-- Found Git: /usr/bin/git (found version 2.39.3)
-- Configuring done
-- Generating done
-- Build files have been written to: /tmp/junk
```



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```
[SES] CM0+ frequency is 100 MHz
[SES] os Kernel 010.4.2
[SES] Main Task - looping forever...
[SRU] RX<- STD= 0x0CE, Receiver ID=4, Address=0x9083FFA8
[TTV] SERVICES version 0.0.6
[TTV] ** TEST heartbeat error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTV] ** TEST pinmux error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTV] ** TEST padcontrol error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTV] ** TEST crypto TRNG 64 error_code=SERVICES_REQ_SUCCESS 64-bit Random value = 0x1cfed9edc1501c20 service_resp=0
[TTV] ** TEST crypto TRNG 32 error_code=SERVICES_REQ_SUCCESS 32-bit Random value = 0xa479c88f service_resp=0
[TTV] ** TEST crypto TRNG 64 error_code=SERVICES_REQ_SUCCESS 64-bit Random value = 0xhc3fcf15f8479420 service_resp=0
[TTV] ** TEST crypto TRNG 32 error_code=SERVICES_REQ_SUCCESS 32-bit Random value = 0xScfbb70c service_resp=0
[TTV] ** TEST crypto LCS error_code=SERVICES_REQ_SUCCESS LCS State 0x0 service_resp=0
[TTV] ** TEST get ATOC error_code=SERVICES_REQ_SUCCESS Application TOC number = 0 service_resp=0x00000000
[TTV] ** TEST TOC via name HE error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTV] ** TEST TOC via name HP error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
[TTV] ** TEST TOC via cpuid error_code=SERVICES_REQ_SUCCESS id HE_DBG flags 63 1 0 0 service_resp=0x00000000
[TTV] ** TEST TOC via cpuid error_code=SERVICES_REQ_SUCCESS id HE_DBG flags 63 1 0 0 service_resp=0x00000000
[TTV] ** TEST soc id error_code=SERVICES_REQ_SUCCESS Device number 0xA100 service_resp=0x00000000
[TTV] ** TEST Boot TOC A32 error_code=SERVICES_REQ_SUCCESS service_resp=0x00000000
:
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services-cmsis.rteconfig main.c services_lib_interface.c M55_HE.sct

Components ☒ Resolve

Software Components	Sel.	Variant	Vendor	Version	Description
AE722F80F55D5AE:M55_H			Alif Semiconductor		ARM Cortex-M55 160 MHz, 13824 KB RAM, 5632 KB ROM
▼ CMSIS					Cortex Microcontroller Software Interface Components
CORE	☒		ARM	5.5.0	CMSIS-CORE for Cortex-M, SC000, SC300, ARMv8-M, ARMv8.1-M
DSP	☐	Source	ARM	1.9.0-dev	CMSIS-DSP Library for Cortex-M, SC000, and SC300
NN Lib	☐		ARM	3.0.0	CMSIS-NN Neural Network Library
> RTOS (API)				1.0.0	CMSIS-RTOS API for Cortex-M, SC000, and SC300
> RTOS2 (API)				2.1.3	CMSIS-RTOS API for Cortex-M, SC000, and SC300
> CMSIS Driver					Unified Device Drivers compliant to CMSIS-Driver Specifications
> Device					Startup, System Setup
Startup	☒	C Startup	AlifSemiconductor	1.0.0	System and Startup for M55_HE device
> RTOS		FreeRTOS	ARM	10.3.1.1	FreeRTOS Real Time Kernel

Validation Output	Description
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Components ☒ Resolve

Software Components	Sel.	Variant	Vendor	Version	Description
AE722F80F55D5AE:M55_H			Alif Semiconductor		ARM Cortex-M55 160 MHz, 13824 KB RAM, 5632 KB ROM
> BSP					
> CMSIS					Cortex Microcontroller Software Interface Components
> CMSIS Driver					Unified Device Drivers compliant to CMSIS-Driver Specifications
> Device					Startup, System Setup
> OSPI XIP					
> SE Services					
MHU Driver	☒		AlifSemiconductor	0.63.0	Message Handling Unit driver for Alif Soc
SE RunTime services	☒	Lib	AlifSemiconductor	0.63.0	SE runtime Services library for RTSS cores
> SOC Peripherals					
Startup	☒	C Startup	AlifSemiconductor	1.0.0	System and Startup for M55_HE device
> FreeRTOS					
> RTOS		AzureRTOS	AlifSemiconductor	0.2.1	Alif Semiconductor port of AzureRTOS for its M55 device

Validation Output	Description
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l r c ffc fft c

A ffffffftfft

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H

typedef struct

```
{
    service_header_t header;
    volatile uint32_t send_<param>;    /*!< Send    parameter */
    volatile uint32_t resp_<param>;    /*!< Return parameter */
    volatile uint32_t resp_error_code; /*!< Call error code */
} example_service_t;
```

void SERVICES example_call(services_req_t *service)

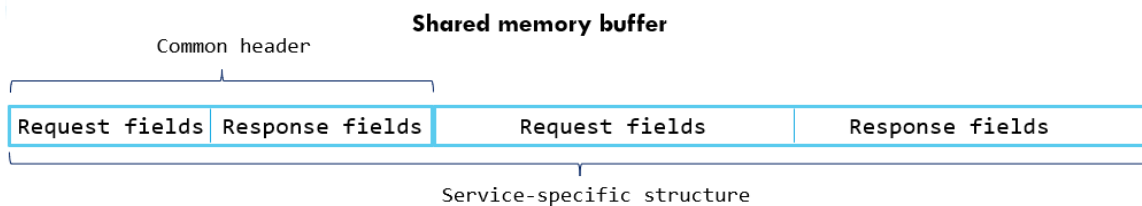
```
{
    example_service_t *p_svc =
        (example_service_t *)service->pkt buffer address; /* services request */
    uint32_t error_code;
    uint32_t local result;

    error_code = function call(p_svc->send_<param>, &local result);

    p_svc->resp_<param>    = local result;
    p_svc->resp_error_code = error_code;

    SERVICES send response code(service, SERVICES_REQ_SUCCESS);
}
```

l r c fct fft l r lrlal cr fft



Common header format

Service ID: req
Flags: req
Error Code: resp
Padding

```
typedef struct
{
    uint16_t send_sid;
    uint16_t send_flags;
    uint16_t resp_error_code; // transport
    layer error code
    uint16_t none_padding;
} service_header_t;
```

Service-specific structure example

```
typedef struct
{
    service_header_t header;
    uint8_t send_port_num;
    uint8_t send_pin_num;
    uint8_t send_config_data;
    uint8_t resp_error_code; // service-specific
    error code
} pinmux_svc_t;
```

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l r c fct fft l fft

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#define
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#define
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#define

l r c fct fft c l ft fft

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static _____

__attribute__

int SERVICES_print const char

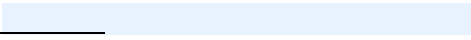
return_____

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_____ **SERVICES wait ms**

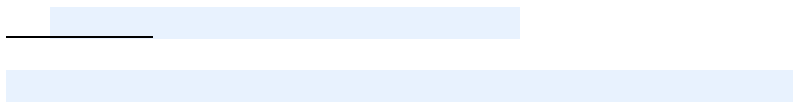
return

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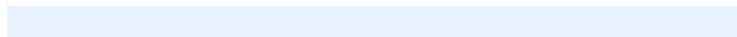


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#define

#define

#define



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A ftc r c r

if

return

return

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int



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rc

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SERVICES system set services debug

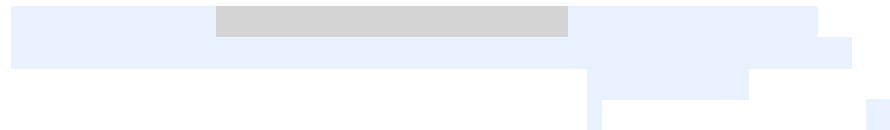
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SERVICES system read otp



if

return

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Arc4

SERVICES_system_write_otp

Arc4

SERVICES_system_get_otp_data

if

return

A rc cr rl a r

SERVICES_system_get_toc_data

typedef struct

typedef struct

H

H

if

return

```
[TTY] ** TEST TOC get data      error_code=SERVICES_REQ_SUCCESS      TOC number = 5 service_resp=0x00000000
[TTY] +-----+
[TTY] |  Name   |  CPU   | Load Address | Boot Address | Image Size | Version | Flags |
[TTY] +-----+
[TTY] |  DEVICE | CM0+   | 0x00000000   | 0x00000000   |    296    | 0.5.0 | u v |
[TTY] |  DEVICE | CM0+   | 0x00000000   | 0x00000000   |    372    | 0.5.0 | u v |
[TTY] |  SERAM0 | CM0+   | 0x00000000   | 0x00000000   |   57612   | 1.105.0 | u s |
[TTY] |  SERAM1 | CM0+   | 0x00000000   | 0x00000000   |   57612   | 1.105.0 | u s |
[TTY] |  SRV-HE-T | M55_HE | 0x58000000   | 0x58000000   |   73440   | 1.0.0 | uLVB |
[TTY] +-----+
```

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SERVICES_system_get_toc_number

```
if
return
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Arc rla c fft

_____ SERVICES system get toc version _____

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Arc rla ft c

SERVICES_system_get_toc_via_name
const

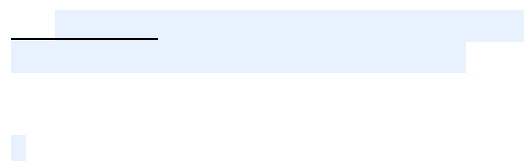
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Arc rla fft a fft

_____ SERVICES_system_get_toc_via_cpuid _____

typedef enum



if

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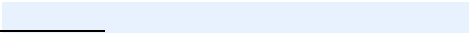
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A rc cr c ffc r c

_____ SERVICES_system_get_device_part_number _____

0x0000B200

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if

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Arc cr c fac r

SERVICES_system_get_device_data

```
typedef struct
```

[illegible]

Government	Percentage
Current government	75%
Previous government	25%

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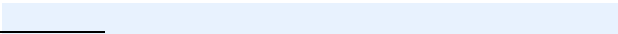
return

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A cr c c fffft

_____ SERVICES_get_se_revision _____

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A rc cr c ffc rc fft

SERVICES system get eui extension

A rc cr c ffc fft

SERVICES system get device id64

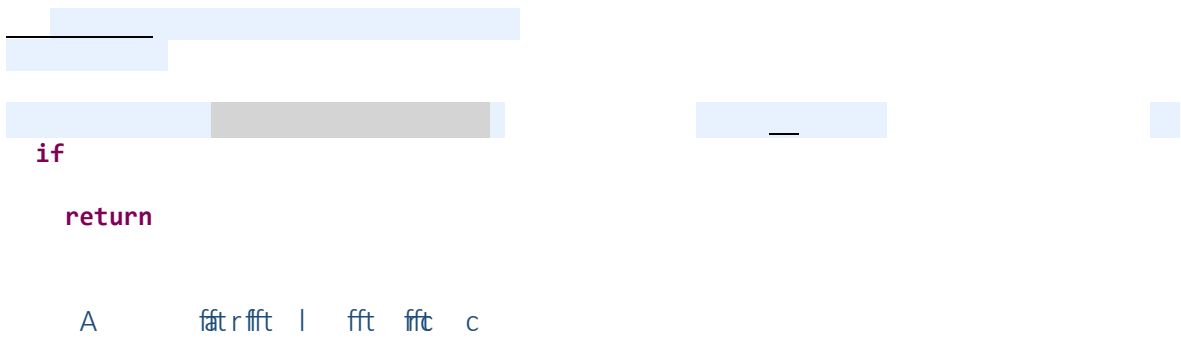
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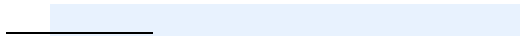
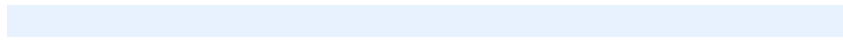


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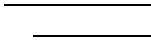
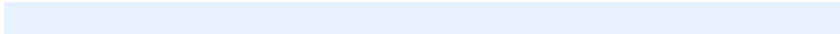
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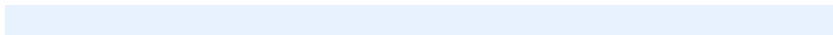


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typedef enum



typedef enum



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#define
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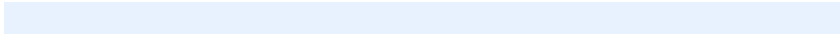
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#define
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typedef enum
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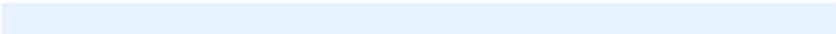


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typedef enum



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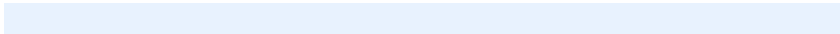
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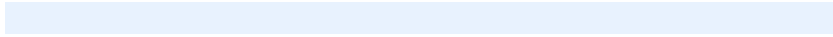
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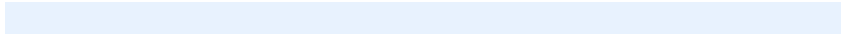
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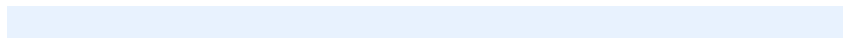
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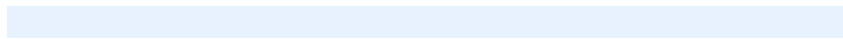
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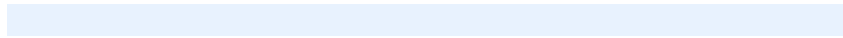
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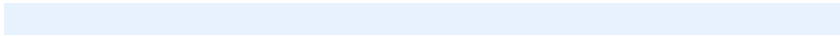


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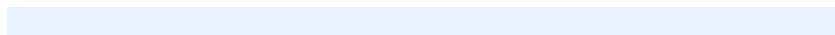
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typedef enum

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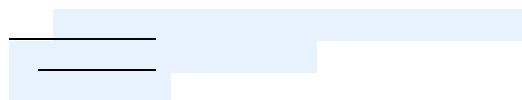
A l l r c c r l a

H
H



A r l c ffc

A a r l a c c r



H
H

if

return

A a rl ac cr a



if

return

H
H

SERVICES_cryptocell_mbedtlsls_hardware_poll

H

H
H



SERVICES_cryptocell_mbedtls_aes_init

SERVICES_cryptocell_mbedtls_aes_set_key

SERVICES_cryptocell_mbedtls_aes_crypt



SERVICES_cryptocell_mbedtls_sha_starts

H
H



SERVICES_cryptocell_mbedtls_sha_process

SERVICES_cryptocell_mbedtls_sha_update

SERVICES_cryptocell_mbedtls_sha_finish

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H



SERVICES_cryptocell_mbedtls_ccm_gcm_set_key

SERVICES_cryptocell_mbedtls_ccm_gcm_crypt

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SERVICES_cryptocell_mbedtls_chacha20_crypt

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H



SERVICES_cryptocell_mbedtls_chachapoly_crypt

SERVICES_cryptocell_mbedtls_poly1305_crypt

SERVICES_cryptocell_mbedtls_cmac_init_setkey

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SERVICES_cryptocell_mbedtls_cmac_update

SERVICES_cryptocell_mbedtls_cmac_finish

SERVICES_cryptocell_mbedtls_cmac_reset

H
H



R r c c f t

A r c r l a

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H

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H
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H
H



H
H



A f t r l c r l a

H
H



l a c r ffitl

H
H